

UP crew

Flies like a rotten place to lay their eggs.

The clean-up crew



bodies? The answer lies in a group of organisms whose way of life depends on the death of others. These are *nature's clean-up crew*—the scavengers and decomposers.



DECOMPOSERS

Decomposers are a vital part of the food chain because they turn organic materials back into simple chemical nutrients, such as carbon, nitrogen, and oxygen. These are then released back into the air, soil, or water. It's easy to tell when decomposers are at work—they are responsible for many of the smells of rotting food and the slimy goo that is left behind.

CHEMICAL CONVERTERS



Fungi

Fungi and molds can't make or catch their own food—instead they grow on dead plants and animals. They send out root-like structures (hyphae) that secrete enzymes that turn dead matter into nutrients.



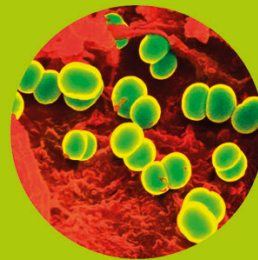
Molds

Molds are a type of fungus that grow in colonies on rotting food. They spread across the surface in a network of white hyphae called a mycelium. Molds reproduce using gray, green, or brown spores.



Insects

Insects are vital decomposers. Many lay their eggs in rotting material, which hatch into maggots and grubs. These feed on the material, stripping carcasses to the bone and returning nutrients to the soil.



Bacteria

Bacteria are everywhere. Ninety percent of the microorganisms in a gram of soil are bacteria and between them they will digest almost anything. They bring about the final breakdown of substances.



The problem with poop...

It isn't only dead bodies that need to be disposed of. Animals can't digest everything they eat, so they have to excrete it as waste along with dead cells from their own body. Luckily, one animal's poop is another organism's dinner, which is why we're not up to our eyeballs in the stuff. In fact, decomposers delight in animal waste because it is partially broken down and contains vital nutrients.

Dung lunch

Dung beetles eat, live, and breed in animal droppings and will even steal a precious ball of dung from another beetle. Some ride along on an animal, waiting for it to poop.



Sewage to fertilizer

Water and sewage treatment works take advantage of bacteria's taste for human waste, using them to turn sewage into clean water and fertilizer that can be spread onto fields.

